THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET A

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2\,NaCl$ is an example of: 1

(a) double displacement (precipitation) **(b)** redox
(c) decomposition **(d)** combination

2. When silver nitrate solution is added to sodium chloride solution, the white precipitate formed is: 1

(a) silver oxide **(b)** sodium chloride **(c)** silver chloride **(d)** sodium nitrate

3. The brown gas evolved on heating lead nitrate is: 1

(a) O_2 **(b)** NO_2 **(c)** N_2 **(d)** NO

4. Tooth decay starts when the pH of the mouth is lower than: 1

(a) 3 **(b)** 5.5 **(c)** 7.5 **(d)** 9

5. Which acid is present in tomato? 1

(a) citric acid **(b)** tartaric acid **(c)** lactic acid **(d)** oxalic acid

6. A student states: “Acids react with all metals to give hydrogen gas.” The correct response is: 1

(a) Correct
(b) Correct for dilute acids
(c) Incorrect — less reactive metals like copper do not give hydrogen
(d) Cannot be determined

7. Which metal is stored under kerosene? 1

(a) sodium **(b)** iron **(c)** aluminium **(d)** copper

8. The best conductors of heat among metals are: 1

(a) lead and mercury **(b)** iron and zinc
(c) silver and copper **(d)** sodium and potassium

9. The non-metal that is a liquid at room temperature is: 1

(a) mercury **(b)** chlorine **(c)** iodine **(d)** bromine

10. Which of these is an allotrope of carbon made of 60 atoms? 1

(a) graphite **(b)** diamond **(c)** charcoal **(d)** fullerene

11. The IUPAC name of CH_3COOH is: 1

(a) methanoic acid **(b)** ethanol **(c)** ethanal **(d)** ethanoic acid

12. The gas used in welding, which burns with a hot flame with oxygen, is: 1

(a) ethene **(b)** ethyne **(c)** ethane **(d)** methane

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Why should a magnesium ribbon be cleaned before burning it in air? 2

14. Why does an aqueous solution of an acid conduct electricity? 2

15. Write the chemical name and formula of Plaster of Paris. 2

16. A student says that hydrogen gas is evolved when any metal reacts with nitric acid. Justify or refute. 2

17. What is meant by tetravalency of carbon? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. An iron nail is dipped in copper sulphate solution for 20 minutes. Write two observations, the equation and the type of reaction. 3

19. Explain the chlor-alkali process with the equation and name the products formed at the anode and cathode. 3

20. What is the thermit reaction? Write its equation and one use. 3

21. What is an addition reaction? Explain with an example. Give one industrial application. 3

22. Draw the electron-dot structures of 3
(i) ethane
(ii) ethene
(iii) carbon dioxide.

OR

What is a homologous series? Give three characteristics, with the example of alkanes. 3

23. Translate into balanced chemical equations: 3
(i) Hydrogen combines with nitrogen to form ammonia.
(ii) Hydrogen sulphide burns in air to give water and sulphur dioxide.
(iii) Barium chloride reacts with aluminium sulphate to give aluminium chloride and barium sulphate.

OR

Why do we paint iron articles? Why are chips packets flushed with nitrogen? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** Neha was stung by a honey-bee. Her mother applied baking soda on the sting. 5
(i) Name the acid injected by the bee.
(ii) Why was baking soda applied?
(iii) Name the type of reaction.
(iv) What is the pH of baking soda solution — less or more than 7?
(v) Suggest one other remedy.

OR

(i) How is Plaster of Paris obtained? Write the equation. 5
(ii) What happens when it is mixed with water?
(iii) Give two uses.
(iv) Why should it be stored in a moisture-proof container?

25. (i) Write three reactions of ethanoic acid with equations (with base, carbonate, alcohol). 5
(ii) Why is ethanoic acid called a weak acid?

OR

(i) Write the structural formulae of all isomers of pentane (C_5H_{12}). 5
(ii) What is isomerism?
(iii) Can methane show isomerism? Why?

END OF QUESTION PAPER — Total: 50 Marks